

AMARJEET KUSHWAHA

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DATA SCIENCE & DATA ENGINEERING | MACHINE LEARNING, NLP & DATA PIPELINES

PROFESSIONAL PROFILE

Data Science MSc graduate with Merit and three years of digital operations experience, delivering end-to-end data and machine-learning workflows. Applied Python, R, SQL, XGBoost, PyTorch and Transformers across fraud detection, NLP and reinforcement learning, alongside ingestion, ETL, relational modelling and data-pipeline delivery; selected models achieved 0.820 PR-AUC and 97.90% F1.

TECHNICAL SKILLS

Programming & Data: Python, R, SQL, pandas, NumPy, Jupyter, Matplotlib, data cleaning, exploratory analysis, feature engineering, statistical analysis

Machine Learning: scikit-learn, XGBoost, ensemble learning, imbalanced classification, cross-validation, class weighting, ROC-AUC, PR-AUC, F1

NLP & Deep Learning: PyTorch, Hugging Face Transformers, DistilBERT, TF-IDF, text classification, tokenisation, adversarial robustness testing

Data Engineering & Delivery: MySQL, SQL Server, ETL pipelines, data migration, 3NF relational modelling, indexing, data validation, Stable-Baselines3, Gymnasium, Git/GitHub

PROFESSIONAL EXPERIENCE

Website Manager | Jashoda Industry

Oct 2021 - Nov 2024

- Used Python and MySQL to analyse user behaviour, contributing to a 25% increase in customer engagement.
- Automated Power BI and Looker Studio dashboards, cutting manual reporting effort by 40%.
- Built real-time ETL pipelines, improving reported data reliability to 95% across website and marketing reporting.
- Managed analytics, SEO, content and performance optimisation, maintained 99.9% uptime and partnered across IT and marketing.

WordPress Developer & SEO Intern | Dipeshpreneur

Jan 2021 - Oct 2021

- Built Excel and Google Analytics dashboards tracking bounce rate, conversion, sessions and campaign performance.
- Analysed web and social metrics to guide content strategy, contributing to 15% higher audience retention.
- Applied keyword, backlink and page-speed analysis, increasing traffic by 20%, visibility by 30% and page speed by 45%.

DATA SCIENCE & ENGINEERING PROJECTS

Phishing Email Classification & Robustness Evaluation

MSc Dissertation | Python | NLP | scikit-learn | PyTorch | Transformers

- Processed 18,650 labelled emails, removed 16 incomplete records, engineered 13 statistical/security indicators and 5,000 TF-IDF features, and balanced classes to 22,644 samples.
- Benchmarked six traditional machine-learning models and fine-tuned DistilBERT; saved evaluation results recorded 97.90% test accuracy, 97.90% F1 and 99.78% ROC-AUC.
- Automated clean-versus-perturbed robustness reporting across multiple strengths; at epsilon 0.10, recorded accuracy fell 0.02 percentage points for DistilBERT versus 26.14 for Random Forest.
- Produced reusable CSV result tables, ROC curves, confusion matrices and robustness charts for comparative reporting.

Transaction Fraud Detection & Risk Modelling

R | Python | XGBoost | scikit-learn | Imbalanced classification

- Prepared 299,695 e-commerce transactions in R for a 2.21% fraud class, validating quality and engineering behavioural and cyclical time features across 21 predictors.
- Developed a weighted-average ensemble and XGBoost; applied a stratified 80/20 split and 44.32 class weighting, then compared four categorical encoders.
- Achieved 0.974 ROC-AUC and 0.820 PR-AUC, detecting 1,150 of 1,322 fraud cases (87.0% recall) across 59,939 test transactions.
- Identified actionable risk signals: failed AVS checks showed 9.67% fraud versus 0.76% when matched, while median shipping distance was 626 km for fraud versus 269 km otherwise.

Relational Database Migration & SQL Optimisation

SQL Server | 3NF data modelling | Data migration | Indexing | BI-ready datasets

- Peer-reviewed and finalised a nine-table 3NF SQL Server schema separating customer/location, product/variant and order entities with PK/FK and uniqueness controls.
- Aggregated 46,041 legacy line items into 16,787 OrderGroup records, deriving line counts and totals while reconciling 2,789,946.60 in transaction value exactly.
- Designed 10 targeted non-clustered indexes across customer, order, product and variant join/filter columns to support analytical queries and downstream reporting.
- Prepared and handed off the normalised SQL dataset for downstream SSAS cube analysis and BI reporting.

Deep Reinforcement Learning for Autonomous Lunar Landing

Python | Stable-Baselines3 | Gymnasium | PyTorch | DQN | PPO

- Developed DQN and PPO agents for Gymnasium LunarLander-v3 using Stable-Baselines3 and PyTorch, learning control policies from an eight-dimensional state space and four discrete actions.
- Designed a controlled benchmark with 300,000 training timesteps per agent and 100 seeded test episodes, comparing learned policies with a random baseline.
- Tuned DQN experience replay, exploration and a 256-by-256 neural network, then produced learning-curve, reward and outcome visualisations to assess convergence and sample efficiency.
- Achieved a 152.9 mean evaluation reward and 53% landing success with DQN, outperforming PPO at 105.1 mean reward and 17% success under the same training budget.

EDUCATION

MSc Data Science (Information Technology), Merit

Feb 2025 - Jan 2026

University of Surrey

B.Tech Computer Science & Engineering | GPA: 7.86

Aug 2017 - May 2021

ITM Vocational University

ADDITIONAL UK EXPERIENCE

Crew Member | McDonald's, New Malden

Feb 2026 - Present

Handle orders and payments accurately, uphold food-safety standards and resolve issues during busy service.

Customer Support | Franco Manca, Guildford

Nov 2025 - Feb 2026

Resolved customer queries and supported orders, payments, bookings and special requests during busy shifts.